

ONON, Mongolia

WMO: 442570

Lat: 48.6161N

Lon: 110.5989E

Elev: 2181

StdP: 77.72

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.0	-31.9	-37.6	0.1	-33.0	-35.6	0.2	-30.8	11.8	-16.8	10.3	-17.4	0.3	320	0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	30.5	15.7	28.1	15.1	26.1	14.7	18.2	25.5	17.3	24.2	16.4	22.9	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	15.1	20.9	15.1	14.1	20.2	14.1	13.2	19.4	61.3	25.7	57.9	23.8	54.7	23.0	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.7	8.7	DB	-36.6	33.7	2.3	2.5	-38.2	35.5	-39.6	36.9	-40.8	38.3	-42.5	40.1	
(5)			WB	-36.7	20.3	2.2	1.3	-38.3	21.2	-39.6	22.0	-40.8	22.7	-42.4	23.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.0	-21.4	-16.8	-6.9	2.6	9.7	16.3	18.9	16.7	10.2	0.9	-11.6	-19.7	(6)	
(7)		DBStd	15.01	5.07	5.63	6.47	4.95	4.38	3.49	2.77	3.44	4.13	4.73	6.66	5.87	(7)	
(8)		HDD10.0	4428	973	750	523	225	59	2	0	1	47	283	647	919	(8)	
(9)		HDD18.3	6772	1231	983	781	471	269	79	24	72	246	541	897	1178	(9)	
(10)		CDD10.0	781	0	0	0	4	50	190	276	209	52	1	0	0	(10)	
(11)		CDD18.3	83	0	0	0	0	2	18	42	21	1	0	0	0	(11)	
(12)		CDH23.3	1222	0	0	0	2	82	346	463	296	34	0	0	0	(12)	
(13)		CDH26.7	376	0	0	0	0	23	113	151	83	6	0	0	0	(13)	
(14)	Wind	WSAvg	3.0	2.3	2.9	3.2	4.0	4.4	3.2	2.6	2.7	3.1	2.9	2.6	2.5	(14)	
(15)	Precipitation	PrecAvg	328	2	3	5	9	20	59	97	71	37	13	5	3	(15)	
(16)		PrecMax	518	6	8	23	30	86	129	233	244	104	46	15	7	(16)	
(17)		PrecMin	158	0	0	0	0	1	8	32	24	1	0	0	0	(17)	
(18)		PrecStd	93	2	2	5	7	18	35	48	48	25	10	4	2	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.8	2.4	14.1	21.7	29.8	33.3	34.3	31.8	27.5	18.9	6.9	-2.6	(19)	
(20)			MCWB	-8.9	-3.0	3.4	7.7	12.2	15.6	17.8	17.1	13.7	7.8	0.8	-5.7	(20)	
(21)		2%	DB	-9.0	-1.2	9.8	18.1	25.7	29.9	30.6	29.0	24.0	15.3	3.1	-6.1	(21)	
(22)			MCWB	-11.3	-5.2	1.6	6.1	10.4	14.8	16.8	15.9	11.8	5.6	-1.3	-8.3	(22)	
(23)		5%	DB	-11.4	-4.2	6.4	15.0	22.8	27.0	27.7	26.6	21.3	12.2	0.7	-8.4	(23)	
(24)			MCWB	-13.1	-7.3	-0.2	4.7	9.6	13.8	16.3	15.3	11.2	4.0	-2.9	-10.2	(24)	
(25)		10%	DB	-13.4	-7.0	3.1	12.1	20.0	24.8	25.6	24.4	18.9	9.3	-1.6	-10.9	(25)	
(26)			MCWB	-14.8	-9.5	-2.1	3.5	8.5	13.3	15.9	14.8	10.2	2.7	-4.6	-12.3	(26)	
(27)		Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.7	-2.7	3.8	8.7	13.4	17.3	20.2	19.0	15.0	8.2	1.0	-5.5	(27)
(28)				MCDWB	-5.9	2.3	13.1	19.2	26.7	28.4	27.6	27.9	23.4	17.9	6.3	-2.7	(28)
(29)	2%		WB	-11.2	-5.1	1.7	6.7	11.5	16.1	18.7	17.7	13.7	6.0	-1.0	-8.3	(29)	
(30)			MCDWB	-9.1	-1.3	9.3	16.6	23.0	24.5	25.8	24.7	20.9	14.3	2.7	-6.2	(30)	
(31)	5%		WB	-13.1	-7.4	0.0	5.2	10.2	15.3	17.9	16.7	12.6	4.4	-2.7	-10.3	(31)	
(32)			MCDWB	-11.4	-4.3	6.2	14.2	21.1	23.1	24.7	23.4	18.8	11.3	0.5	-8.5	(32)	
(33)	10%		WB	-14.8	-9.4	-1.8	3.8	8.9	14.4	17.1	15.7	11.2	3.0	-4.5	-12.3	(33)	
(34)			MCDWB	-13.5	-7.1	2.8	11.6	18.8	22.0	23.4	22.0	17.5	8.8	-1.7	-11.0	(34)	
(35)	Mean Daily Temperature Range		5% DB	MDBR	11.4	14.0	14.2	13.7	14.9	13.6	11.9	12.9	14.1	13.0	11.6	10.3	(35)
(36)				MCDBR	13.5	17.7	18.2	19.6	21.2	18.9	16.4	17.5	19.2	18.5	13.8	12.3	(36)
(37)		MCWBR		11.7	14.0	11.3	10.2	9.4	6.8	5.7	6.6	9.3	10.2	10.0	10.6	(37)	
(38)		5% WB	MCDWB	13.4	17.5	17.7	17.5	18.3	14.0	12.8	13.4	15.4	16.5	13.6	12.3	(38)	
(39)			MCWBR	11.7	13.9	11.1	9.3	8.7	5.9	5.5	5.7	7.9	9.4	10.0	10.6	(39)	
(40)																	(40)
(41)	Clear-Sky Solar Irradiance	taub	0.205	0.222	0.251	0.293	0.333	0.341	0.321	0.301	0.270	0.242	0.218	0.205	(41)		
(42)		taud	2.448	2.443	2.453	2.404	2.310	2.378	2.480	2.520	2.564	2.576	2.510	2.438	(42)		
(43)		Ebn at Noon	921	971	983	961	926	919	934	942	949	935	894	873	(43)		
		Edh at Noon	59	76	91	107	123	116	103	95	81	67	55	52			
(44)	All-Sky Solar Radiation	RadAvg	1.85	3.04	4.56	5.61	6.33	6.25	5.93	5.41	4.41	3.07	1.95	1.46	(44)		
(45)		RadStd	0.08	0.15	0.19	0.27	0.34	0.38	0.32	0.32	0.32	0.14	0.09	0.12	(45)		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (4 neighbors)		N/A	N/A	-1.25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page